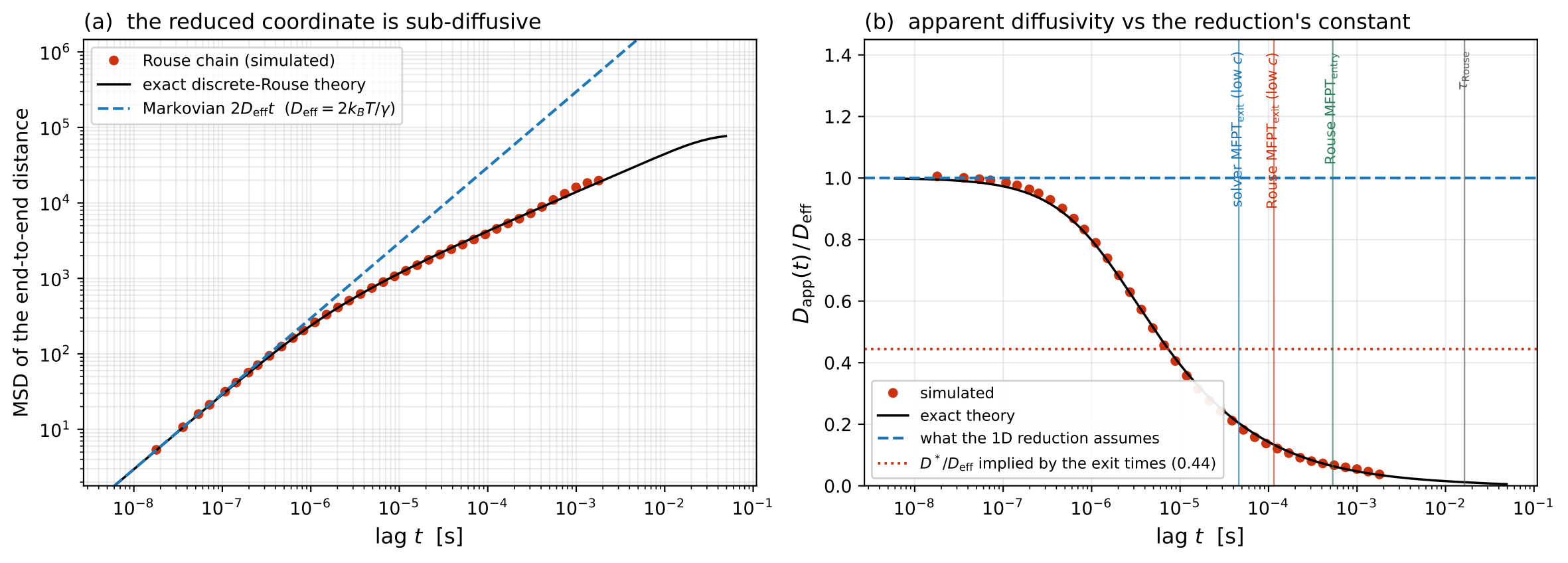
(a) the reduced coordinate is sub-diffusive
Rouse chain (simulated)
exact discrete-Rouse theory
Markovian $2D_{\mathrm{eff}}t$ ($D_{\mathrm{eff}} = 2k_BT/\gamma$)
MSD of the end-to-end distance
lag $t$ [s]
(b) apparent diffusivity vs the reduction's constant
simulated
exact theory
what the 1D reduction assumes
$D^*/D_{\mathrm{eff}}$ implied by the exit times (0.44)
solver MFPT$_{\mathrm{exit}}$ (low $c$)
Rouse MFPT$_{\mathrm{exit}}$ (low $c$)
Rouse MFPT$_{\mathrm{entry}}$
$\tau_{\mathrm{Rouse}}$
$D_{\mathrm{app}}(t)/D_{\mathrm{eff}}$
lag $t$ [s]